

Getting started with CI

SETTING UP A BASIC GITHUB WORKFLOW

Pre-Requisites

Virtual Machine with Windows

Required developement environments and tools

Some examples:

- LabVIEW
- VIPM tools
- TestStand
- Python
- PowerShell Core (pwsh)

Be mindfull of versions and bitness

GitHub Action Runner Setup

The screenshot shows the GitHub Actions Runner setup page for the Astemes organization. The browser address bar displays `github.com/organizations/astemes/settings/actions/runners/new`. The page header includes the Astemes organization name, a search bar, and navigation links for Overview, Repositories (119), Projects, Packages, Teams (1), People (2), Insights, and Settings. The left sidebar lists various settings categories: General, Policies, Access, Billing and licensing, Organization roles, Repository roles, Member privileges, Import/Export, and Moderation. The main content area is titled "Runners / Create self-hosted runner" and contains a paragraph explaining the requirements for adding a self-hosted runner, including a link to the GitHub Customer Agreement. Below this, the "Runner image" section shows three options: macOS, Linux, and Windows, with Windows selected. The "Architecture" section shows a dropdown menu with "x64" selected.

github.com/organizations/astemes/settings/actions/runners/new

astemes

Overview Repositories 119 Projects Packages Teams 1 People 2 Insights Settings

Astemes Organization

Go to settings page Go to your organization profile

General Policies Access Billing and licensing Organization roles Repository roles Member privileges Import/Export Moderation

Runners / Create self-hosted runner

Adding a self-hosted runner requires that you download, configure, and execute the GitHub Actions Runner. If you do not already have an existing volume licensing agreement for your GitHub purchases, by downloading and configuring the GitHub Actions Runner, you agree to the [GitHub Customer Agreement](#).

Runner image

macOS Linux Windows

Architecture

x64

Follow the instructions

Download

We recommend configuring the runner under "actions-runner". This will help avoid issues related to service identity folder permissions and long path restrictions on Windows.

```
# Create a folder under the drive root
$ mkdir actions-runner; cd actions-runner

# Download the latest runner package
$ Invoke-WebRequest -Uri https://github.com/actions/runner/releases/download/v2.334.0/actions-runner-win-x64-2.334.0.zip -OutFile actions-runner-win-x64-2.334.0.zip

# Optional: Validate the hash
$ if((Get-FileHash -Path actions-runner-win-x64-2.334.0.zip -Algorithm SHA256).Hash.ToUpper() -ne 'a0c9e5f5-f87d11-47facc-8ccc1b8cc1-75f6cccc11725f07-ee-56cc11-7411-(*)5f11-1a-11111111111111111111')
```



Configure

```
# E
$ A
win
# Create the runner and start the configuration experience
$ ./config.cmd --url https://github.com/astemes --token
# Run it!
$ ./run.cmd
```

Preparations

Installed

- LabVIEW
- LabVIEW Command Line Interface
- VIPM
- LUnit
- LUnit CLI
- Git for windows

Configured GitHub runner with label "nordlug"

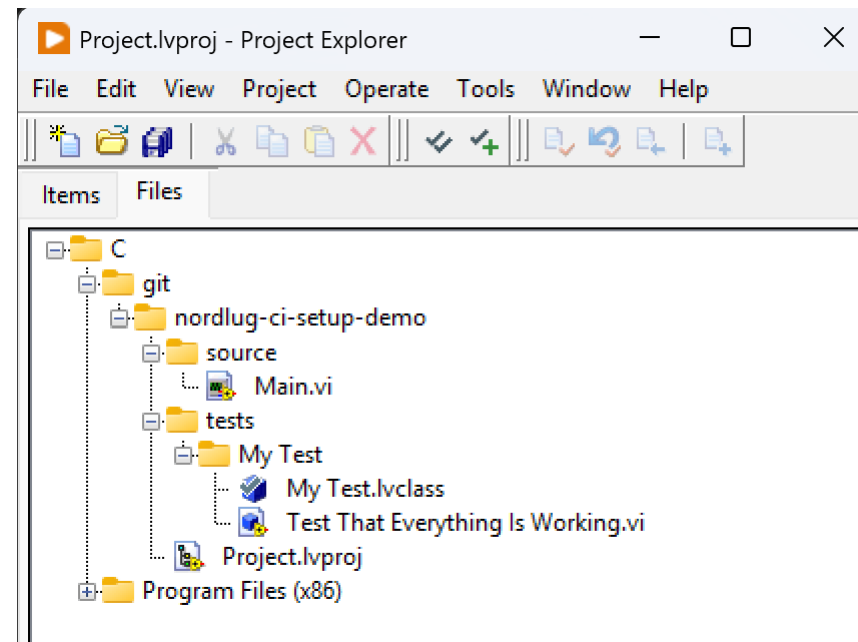
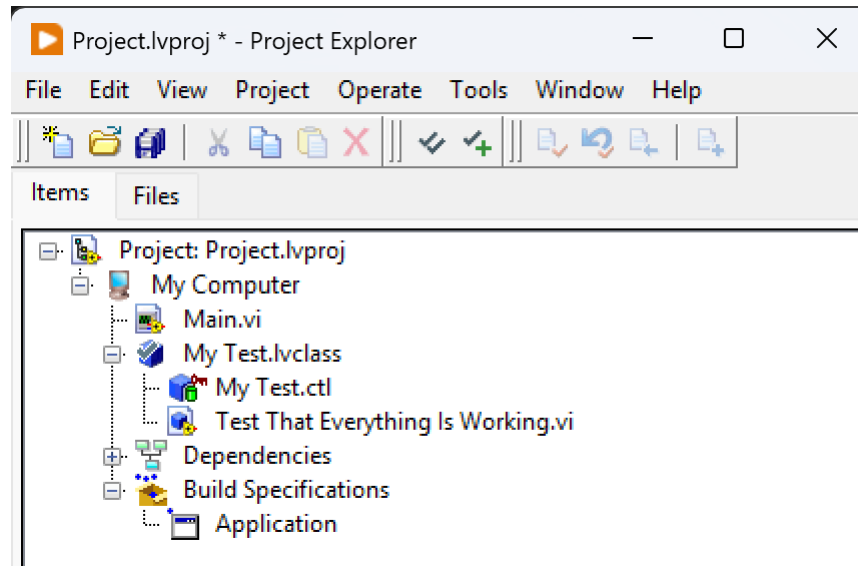
Configured the environment variables (used by scripts in *astemes-github-actions*):

- \$env:LabVIEWPath = "C:\Program Files\National Instruments\LabVIEW 2026\LabVIEW.exe"
- \$env:LabVIEWPort = "3363"

Set powershell execution policy to allow scripts

- *Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope LocalMachine -Force*

Project Overview



Demo time!

PRICES TO THE ONES WHO SPOTS THE INTENTIONAL MISTAKES